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Editors
General Relativity and Gravitation

Dear Editors,

Please consider my manuscript, “A Strict Surface-Energy Obstruction for Timelike Kantowski–Sachs–Schwarzschild Junctions,” for publication in *General Relativity and Gravitation*. The manuscript extracts an invariant sufficient condition for negative Israel surface density: under the ordinary orientation, a timelike spherical shell joining a sector with $(\nabla R)^2 \leq 0$ to one with $(\nabla R)^2 > 0$ must have $\sigma < 0$. It then proves the strict Kantowski–Sachs/Schwarzschild realization: throughout the $F > 0$ region outside the Schwarzschild horizon, every compatible finite-rapidity junction to the retained asymptotically flat exterior requires negative surface density for either explicitly retained Kantowski–Sachs interval. For an exact static comoving family, the null energy condition additionally fails precisely outside the Schwarzschild photon sphere, with saturation at $R_0 = 3m$, and an explicit Nariai product region supplies a local geometric witness.

The significance of the result is that it separates an invariant obstruction caused by the Kantowski–Sachs product geometry from coordinate, trajectory, and orientation artifacts. It also supplies a classical boundary diagnostic for proposed black-hole-to-cosmology completions and for effective black-hole interiors formulated in Kantowski–Sachs variables. This combination of exact junction analysis, global branch bookkeeping, and energy-condition classification is well aligned with the scope of *General Relativity and Gravitation*.

The manuscript is original, is not under consideration elsewhere, and has been prepared independently without institutional or grant support. The manuscript source, figure-generation scripts, and verification suite are publicly available at github.com/RonBibb/junction-ec-paper.

Subject to the journal’s conflict-of-interest checks, potential referees with relevant expertise include João L. Rosa, Sebastian Carloni, S. H. Mazharimousavi, José M. M. Senovilla, and S. Khakshournia.

Sincerely,

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